

Firmware Release Notes

DocPlus+ Pulse Oximeter FW-v3.4 Complaint Evidence Refresh

Document ID	FRN-FW34-C100
Firmware Version	FW-v3.4 release family
Release Date	2026-07-01
Device	DocPlus+ Pulse Oximeter
Change Dataset	Firmware_Change_Impact_Assessment_FW-v3.4.xlsx
Release Status	Recommended for release after approval signatures

Document Information

Field	Value
Document title	DocPlus+ Pulse Oximeter Firmware Release Notes FW-v3.4
Document number	FRN-FW34-C100
Release date	2026-07-01
Firmware versions represented	FW-v3.3.8, FW-v3.4.2, FW-v3.4.9, FW-v3.5.3, FW-v3.6.0, FW-v3.3.4, FW-v3.4.0, FW-v3.4.6, FW-v3.5.1, FW-v3.5.7, FW-v3.2.9
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Quality record	Controlled firmware change-control and release evidence summary.

Purpose

These release notes summarize the firmware change impact assessment for 100 complaint-driven firmware change items. The notes identify modules affected, safety and risk impact, verification coverage, regression coverage, deployment considerations, and final release recommendation.

Scope

Scope includes 100 firmware change items across the following modules: power_manager, display_task, measurement_core, optical_frontend, alarm_manager, storage_manager, connectivity_stack, update_identity_manager, system_supervisor, mechanical_usability. The assessment covers software functionality, patient safety, alarms, measurements, user interface, data integrity, communications, performance, cybersecurity, usability, and regulatory traceability where applicable.

Modules Affected

Module	Complaint Count	Critical/Major Count	Representative Firmware Versions	Primary Verification Focus
power_manager	10	10	FW-v3.3.8, FW-v3.4.2, FW-v3.4.9, FW-v3.5.3, FW-v3.6.0, FW-v3.3.4, FW-v3.4.0, FW-v3.4.6, FW-v3.5.1, FW-v3.5.7	TC-PWR-001, TC-PWR-002, TC-PWR-003, TC-PWR-004, TC-PWR-005, TC-PWR-006, TC-PWR-007, TC-PWR-008, TC-PWR-009, TC-PWR-010
display_task	10	9	FW-v3.2.9, FW-v3.3.8, FW-v3.4.2, FW-v3.4.9, FW-v3.5.3, FW-v3.6.0, FW-v3.3.4, FW-v3.4.0, FW-v3.4.6, FW-v3.5.1	TC-DISP-011, TC-DISP-012, TC-DISP-013, TC-DISP-014, TC-DISP-015, TC-DISP-016, TC-DISP-017, TC-DISP-018, TC-DISP-019, TC-DISP-020
measurement_core	21	21	FW-v3.5.7, FW-v3.2.9, FW-v3.3.8, FW-v3.4.2, FW-v3.4.9, FW-v3.5.3, FW-v3.6.0, FW-v3.3.4, FW-v3.4.0, FW-v3.4.6, FW-v3.5.1	TC-MEAS-021, TC-MEAS-022, TC-MEAS-023, TC-MEAS-024, TC-MEAS-025, TC-MEAS-026, TC-MEAS-027, TC-MEAS-028, TC-MEAS-029, TC-MEAS-030, TC-MEAS-031, TC-MEAS-032, TC-MEAS-033, TC-MEAS-034, TC-MEAS-035, TC-MEAS-036, TC-MEAS-037, TC-MEAS-038, TC-MEAS-039, TC-MEAS-040, TC-MEAS-041
optical_frontend	9	9	FW-v3.5.1, FW-v3.5.7, FW-v3.2.9, FW-v3.3.8, FW-v3.4.2, FW-v3.4.9, FW-v3.5.3, FW-v3.6.0, FW-v3.3.4	TC-SENS-042, TC-SENS-043, TC-SENS-044, TC-SENS-045, TC-SENS-046, TC-SENS-047, TC-SENS-048, TC-SENS-049, TC-SENS-050

Module	Complaint Count	Critical/Major Count	Representative Firmware Versions	Primary Verification Focus
alarm_manager	10	10	FW-v3.4.0, FW-v3.4.6, FW-v3.5.1, FW-v3.5.7, FW-v3.2.9, FW-v3.3.8, FW-v3.4.2, FW-v3.4.9, FW-v3.5.3, FW-v3.6.0	TC-ALRM-051, TC-ALRM-052, TC-ALRM-053, TC-ALRM-054, TC-ALRM-055, TC-ALRM-056, TC-ALRM-057, TC-ALRM-058, TC-ALRM-059, TC-ALRM-060
storage_manager	9	8	FW-v3.3.4, FW-v3.4.0, FW-v3.4.6, FW-v3.5.1, FW-v3.5.7, FW-v3.2.9, FW-v3.3.8, FW-v3.4.2, FW-v3.4.9	TC-DATA-061, TC-DATA-062, TC-DATA-063, TC-DATA-064, TC-DATA-065, TC-DATA-066, TC-DATA-067, TC-DATA-068, TC-DATA-069
connectivity_stack	6	6	FW-v3.5.3, FW-v3.6.0, FW-v3.3.4, FW-v3.4.0, FW-v3.4.6, FW-v3.5.1	TC-CONN-070, TC-CONN-071, TC-CONN-072, TC-CONN-073, TC-CONN-074, TC-CONN-075
update_identity_manager	6	6	FW-v3.5.7, FW-v3.2.9, FW-v3.3.8, FW-v3.4.2, FW-v3.4.9, FW-v3.5.3	TC-FWID-076, TC-FWID-077, TC-FWID-078, TC-FWID-079, TC-FWID-080, TC-FWID-081
system_supervisor	9	9	FW-v3.6.0, FW-v3.3.4, FW-v3.4.0, FW-v3.4.6, FW-v3.5.1, FW-v3.5.7, FW-v3.2.9, FW-v3.3.8, FW-v3.4.2	TC-SYS-082, TC-SYS-083, TC-SYS-084, TC-SYS-085, TC-SYS-086, TC-SYS-087, TC-SYS-088, TC-SYS-089, TC-SYS-090
mechanical_usability	10	7	FW-v3.4.9, FW-v3.5.3, FW-v3.6.0, FW-v3.3.4, FW-v3.4.0, FW-v3.4.6, FW-v3.5.1, FW-v3.5.7, FW-v3.2.9, FW-v3.3.8	TC-USE-091, TC-USE-092, TC-USE-093, TC-USE-094, TC-USE-095, TC-USE-096, TC-USE-097, TC-USE-098, TC-USE-099, TC-USE-100

Summary of Implemented Changes

The release contains complaint-specific corrective and preventive firmware changes or firmware impact confirmations for power management, display/UI behavior, measurement processing, optical sensor diagnostics, alarm management, storage/data integrity, communications, firmware update/identity, system supervision, and usability-related feedback paths. Each change item has one rewritten impact assessment and one corresponding required verification entry.

Resolved Issues

Disposition	Count
Closed - confirmed device failure	48
Closed - field correction initiated	9
Closed - no fault found	5
Closed - confirmed supplier component issue	6
Closed - confirmed software defect	17
Closed - unable to reproduce after extended test	4
Closed - use/environmental cause	11

Known Limitations

No unresolved failed verification, blocked change item, missing requirement trace, or missing risk-control trace remains in this package. Field-correction and monitoring dispositions remain subject to the post-release effectiveness activities defined in the quality system.

Safety Impact Summary

Safety-impact review confirms that affected change items were evaluated for loss of monitoring, incorrect or stale SpO2/pulse values, missed or delayed alarm annunciation, misleading UI or battery state, data loss, communication/export errors, update/boot recovery failures, and foreseeable use-condition issues. Risk-control verification is required for every change item before release closure.

Risk Assessment Summary

Risk control coverage is 100%. Each change item is linked to a risk ID, a requirement reference, and a verification test case derived from the traceability set. Residual risk remains acceptable only when verification confirms that the change prevents, detects, controls, or clearly notifies the user of the associated failure mode without adverse regression.

Verification Summary

Metric	Result
Impact assessments improved	100
Required verification entries improved	100
Verification coverage	100%
Requirement traceability coverage	100%
Risk control coverage	100%
Duplicate final impact assessments	0
Duplicate final verification entries	0

Regression Testing Summary

Regression testing shall include module-specific unit tests, software integration tests, system verification, risk-control verification, alarm verification, measurement accuracy verification, UI verification, performance/boundary/stress testing, cybersecurity and communication checks where applicable, usability validation for use-condition changes, and final traceability review.

Requirement Traceability Summary

Requirement traceability coverage is 100%. All change items preserve original change IDs and map to existing complaint IDs, requirements, risk controls, and test cases. No non-target workbook columns were changed during the assessment update.

Risk Control Verification Summary

Risk-control verification activities include preventive controls, detection controls, fail-safe behavior, alarm/user notification controls, event logging, data integrity checks, firmware recovery, and regression review for cross-module interfaces. Verification evidence must be retained with the change item, requirement, risk ID, and test case reference.

Deployment Considerations

Deploy under controlled firmware change procedures. Confirm device configuration, firmware image identity, rollback/recovery readiness, service communication, release labeling or IFU updates if applicable, cybersecurity and compatibility review, production/service training needs, and post-release complaint trend monitoring before broad release.

Final Release Recommendation

Release is recommended after approval signatures, provided all required verification records are signed, risk-control evidence is complete, traceability review remains at 100%, and field-correction effectiveness monitoring is retained for applicable dispositions.

Detailed Firmware Change Summary

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-001 CPL-100-001 power_manager FW-v3.3.8	REQ-PWR-001 RISK-PWR-001 TC-PWR-001	Device does not power on. Root cause: Battery protection IC entered permanent ship-mode after marginal USB inrush on an aged cell pack	FWIA-001 (CPL-100-001) impact assessment for power_manager on FW-v3.3.8: the change addresses "Battery protection IC entered permanent ship-mode after marginal USB inrush on an aged cell pack" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC interrupt...	FWIA-001 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-001; verify REQ-PWR-001; confirm...	Closed - confirmed device failure
FWIA-002 CPL-100-002 power_manager FW-v3.4.2	REQ-PWR-002 RISK-PWR-002 TC-PWR-002	Device shuts down unexpectedly during measurement. Root cause: Fuel-gauge learned capacity table was corrupted after brownout during measurement save	FWIA-002 (CPL-100-002) impact assessment for power_manager on FW-v3.4.2: the change addresses "Fuel-gauge learned capacity table was corrupted after brownout during measurement save" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC interrupt handling...	FWIA-002 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-002; verify REQ-PWR-002; confirm...	Closed - confirmed device failure
FWIA-003 CPL-100-003 power_manager FW-v3.4.9	REQ-PWR-003 RISK-PWR-003 TC-PWR-003	Battery drains unusually fast. Root cause: Quiescent current increased because the optical AFE remained awake after failed finger detect exit	FWIA-003 (CPL-100-003) impact assessment for power_manager on FW-v3.4.9: the change addresses "Quiescent current increased because the optical AFE remained awake after failed finger detect exit" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC interrupt...	FWIA-003 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-003; verify REQ-PWR-003; confirm...	Closed - confirmed device failure
FWIA-004 CPL-100-004 power_manager FW-v3.5.3	REQ-PWR-004 RISK-PWR-004 TC-PWR-004	Battery charging indicator does not appear. Root cause: Charge-present GPIO filter missed short charger attach pulses from high-resistance cradle contacts	FWIA-004 (CPL-100-004) impact assessment for power_manager on FW-v3.5.3: the change addresses "Charge-present GPIO filter missed short charger attach pulses from high-resistance cradle contacts" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC interrupt...	FWIA-004 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-004; verify REQ-PWR-004; confirm...	Closed - confirmed device failure
FWIA-005 CPL-100-005 power_manager FW-v3.6.0	REQ-PWR-005 RISK-PWR-005 TC-PWR-005	Battery does not charge fully. Root cause: Cell impedance growth caused charger taper termination before available capacity reached label claim	FWIA-005 (CPL-100-005) impact assessment for power_manager on FW-v3.6.0: the change addresses "Cell impedance growth caused charger taper termination before available capacity reached label claim" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC...	FWIA-005 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-005; verify REQ-PWR-005; confirm...	Closed - confirmed device failure
FWIA-006 CPL-100-006 power_manager FW-v3.3.4	REQ-PWR-006 RISK-PWR-006 TC-PWR-006	Device overheats while charging. Root cause: Charge regulator thermal foldback threshold was exceeded due to non-OEM wall adapter ripple current	FWIA-006 (CPL-100-006) impact assessment for power_manager on FW-v3.3.4: the change addresses "Charge regulator thermal foldback threshold was exceeded due to non-OEM wall adapter ripple current" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC...	FWIA-006 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-006; verify REQ-PWR-006; confirm...	Closed - field correction initiated
FWIA-007 CPL-100-007 power_manager FW-v3.4.0	REQ-PWR-007 RISK-PWR-007 TC-PWR-007	Power button is unresponsive. Root cause: Power key dome switch shifted laterally after repeated edge actuation and no longer closed reliably	FWIA-007 (CPL-100-007) impact assessment for power_manager on FW-v3.4.0: the change addresses "Power key dome switch shifted laterally after repeated edge actuation and no longer closed reliably" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC...	FWIA-007 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-007; verify REQ-PWR-007; confirm...	Closed - confirmed device failure

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-008 CPL-100-008 power_manager FW-v3.4.6	REQ-PWR-008 RISK-PWR-008 TC-PWR-008	Device repeatedly restarts automatically. Root cause: Watchdog reset loop caused by I2C bus lock when the PMIC interrupt line remained asserted	FWIA-008 (CPL-100-008) impact assessment for power_manager on FW-v3.4.6: the change addresses "Watchdog reset loop caused by I2C bus lock when the PMIC interrupt line remained asserted" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC interrupt...	FWIA-008 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-008; verify REQ-PWR-008; confirm...	Closed - confirmed device failure
FWIA-009 CPL-100-009 power_manager FW-v3.5.1	REQ-PWR-009 RISK-PWR-009 TC-PWR-009	Low battery warning appears despite full charge. Root cause: Battery full flag was overwritten by a stale low-voltage alarm persisted in nonvolatile settings	FWIA-009 (CPL-100-009) impact assessment for power_manager on FW-v3.5.1: the change addresses "Battery full flag was overwritten by a stale low-voltage alarm persisted in nonvolatile settings" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC interrupt...	FWIA-009 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-009; verify REQ-PWR-009; confirm...	Closed - confirmed device failure
FWIA-010 CPL-100-010 power_manager FW-v3.5.7	REQ-PWR-010 RISK-PWR-010 TC-PWR-010	Incorrect battery percentage displayed. Root cause: Fuel-gauge sense resistor tolerance at the high end biased remaining-capacity calculation downward	FWIA-010 (CPL-100-010) impact assessment for power_manager on FW-v3.5.7: the change addresses "Fuel-gauge sense resistor tolerance at the high end biased remaining-capacity calculation downward" within the power-management and battery-state firmware. Software impact: evaluate power-state sequencing, charger attach detection, battery fuel-gauge interpretation, PMIC interrupt...	FWIA-010 required verification: perform Unit Testing; Integration Testing; System Testing; Power-path Boundary Testing; Brownout/Recovery Stress Testing; Low-Battery and Thermal Alarm Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-PWR-010; verify REQ-PWR-010; confirm...	Closed - confirmed device failure
FWIA-011 CPL-100-011 display_task FW-v3.2.9	REQ-DISP-011 RISK-DISP-011 TC-DISP-011	OLED/LCD screen remains blank. Root cause: OLED boost converter enable trace fractured at the flex tail bend radius	FWIA-011 (CPL-100-011) impact assessment for display_task on FW-v3.2.9: the change addresses "OLED boost converter enable trace fractured at the flex tail bend radius" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and clinical value/alarm...	FWIA-011 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-011; verify REQ-DISP-011; confirm RISK-DISP-011 risk-control effectiveness and retain...	Closed - confirmed device failure
FWIA-012 CPL-100-012 display_task FW-v3.3.8	REQ-DISP-012 RISK-DISP-012 TC-DISP-012	Display flickers intermittently. Root cause: Display FPC latch relaxed after drop shock, intermittently opening the SPI clock contact	FWIA-012 (CPL-100-012) impact assessment for display_task on FW-v3.3.8: the change addresses "Display FPC latch relaxed after drop shock, intermittently opening the SPI clock contact" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and...	FWIA-012 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-012; verify REQ-DISP-012; confirm RISK-DISP-012 risk-control effectiveness and retain...	Closed - confirmed device failure
FWIA-013 CPL-100-013 display_task FW-v3.4.2	REQ-DISP-013 RISK-DISP-013 TC-DISP-013	Display brightness cannot be adjusted. Root cause: Brightness menu writes were rejected after configuration CRC mismatch in the display preference block	FWIA-013 (CPL-100-013) impact assessment for display_task on FW-v3.4.2: the change addresses "Brightness menu writes were rejected after configuration CRC mismatch in the display preference block" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph...	FWIA-013 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-013; verify REQ-DISP-013; confirm RISK-DISP-013 risk-control effectiveness and retain...	Closed - no fault found
FWIA-014 CPL-100-014 display_task FW-v3.4.9	REQ-DISP-014 RISK-DISP-014 TC-DISP-014	Dead pixels appear on the screen. Root cause: OLED panel lot showed clustered subpixel opens at column-driver bond pads	FWIA-014 (CPL-100-014) impact assessment for display_task on FW-v3.4.9: the change addresses "OLED panel lot showed clustered subpixel opens at column-driver bond pads" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and clinical value/alarm...	FWIA-014 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-014; verify REQ-DISP-014; confirm RISK-DISP-014 risk-control effectiveness and retain...	Closed - confirmed supplier component issue
FWIA-015 CPL-100-015 display_task FW-v3.5.3	REQ-DISP-015 RISK-DISP-015 TC-DISP-015	Screen freezes during operation. Root cause: Display task starved behind alarm rendering during high-rate waveform refresh	FWIA-015 (CPL-100-015) impact assessment for display_task on FW-v3.5.3: the change addresses "Display task starved behind alarm rendering during high-rate waveform refresh" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and clinical...	FWIA-015 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-015; verify REQ-DISP-015; confirm RISK-DISP-015 risk-control effectiveness and retain...	Closed - confirmed software defect

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-016 CPL-100-016 display_task FW-v3.6.0	REQ-DISP-016 RISK-DISP-016 TC-DISP-016	Display shows garbled characters. Root cause: Uninitialized font cache pointer caused invalid glyph lookup after language table switch	FWIA-016 (CPL-100-016) impact assessment for display_task on FW-v3.6.0: the change addresses "Uninitialized font cache pointer caused invalid glyph lookup after language table switch" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and...	FWIA-016 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-016; verify REQ-DISP-016; confirm RISK-DISP-016 risk-control effectiveness and retain...	Closed - confirmed device failure
FWIA-017 CPL-100-017 display_task FW-v3.3.4	REQ-DISP-017 RISK-DISP-017 TC-DISP-017	Display orientation is incorrect. Root cause: Orientation flag was programmed with the wrong default during a limited production rework batch	FWIA-017 (CPL-100-017) impact assessment for display_task on FW-v3.3.4: the change addresses "Orientation flag was programmed with the wrong default during a limited production rework batch" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and...	FWIA-017 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-017; verify REQ-DISP-017; confirm RISK-DISP-017 risk-control effectiveness and retain...	Closed - confirmed device failure
FWIA-018 CPL-100-018 display_task FW-v3.4.0	REQ-DISP-018 RISK-DISP-018 TC-DISP-018	Touch buttons do not respond. Root cause: Capacitive key baseline drifted after disinfectant residue bridged the guard ring	FWIA-018 (CPL-100-018) impact assessment for display_task on FW-v3.4.0: the change addresses "Capacitive key baseline drifted after disinfectant residue bridged the guard ring" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and clinical...	FWIA-018 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-018; verify REQ-DISP-018; confirm RISK-DISP-018 risk-control effectiveness and retain...	Closed - confirmed device failure
FWIA-019 CPL-100-019 display_task FW-v3.4.6	REQ-DISP-019 RISK-DISP-019 TC-DISP-019	Physical navigation buttons stick. Root cause: Tactile navigation plunger developed burrs from mold gate vestige interference	FWIA-019 (CPL-100-019) impact assessment for display_task on FW-v3.4.6: the change addresses "Tactile navigation plunger developed burrs from mold gate vestige interference" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and clinical...	FWIA-019 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-019; verify REQ-DISP-019; confirm RISK-DISP-019 risk-control effectiveness and retain...	Closed - confirmed device failure
FWIA-020 CPL-100-020 display_task FW-v3.5.1	REQ-DISP-020 RISK-DISP-020 TC-DISP-020	Display backlight turns off unexpectedly. Root cause: Backlight PWM driver latched off following ESD strike at the side key bezel	FWIA-020 (CPL-100-020) impact assessment for display_task on FW-v3.5.1: the change addresses "Backlight PWM driver latched off following ESD strike at the side key bezel" within the display and user-interface firmware. Software impact: evaluate display-driver control, frame refresh, brightness/orientation settings, input debounce, font/glyph loading, and clinical...	FWIA-020 required verification: perform Unit Testing; UI Verification; Integration Testing; Display Refresh Stress Testing; Alarm Presentation Verification; Usability Validation; Regression Testing; Traceability Review. Traceability: execute TC-DISP-020; verify REQ-DISP-020; confirm RISK-DISP-020 risk-control effectiveness and retain...	Closed - confirmed device failure
FWIA-021 CPL-100-021 measurement_core FW-v3.5.7	REQ-MEAS-021 RISK-MEAS-021 TC-MEAS-021	SpO2 reading remains at 0%. Root cause: ADC dark-current offset exceeded zero-signal threshold after photodiode leakage increased	FWIA-021 (CPL-100-021) impact assessment for measurement_core on FW-v3.5.7: the change addresses "ADC dark-current offset exceeded zero-signal threshold after photodiode leakage increased" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion...	FWIA-021 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-021; verify REQ-MEAS-021; confirm RISK-MEAS-021...	Closed - confirmed device failure
FWIA-022 CPL-100-022 measurement_core FW-v3.2.9	REQ-MEAS-022 RISK-MEAS-022 TC-MEAS-022	Pulse rate is not detected. Root cause: Pulse detection state machine rejected valid plethysmography because adaptive gain never exited search mode	FWIA-022 (CPL-100-022) impact assessment for measurement_core on FW-v3.2.9: the change addresses "Pulse detection state machine rejected valid plethysmography because adaptive gain never exited search mode" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization...	FWIA-022 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-022; verify REQ-MEAS-022; confirm RISK-MEAS-022...	Closed - confirmed device failure
FWIA-023 CPL-100-023 measurement_core FW-v3.3.8	REQ-MEAS-023 RISK-MEAS-023 TC-MEAS-023	SpO2 reading fluctuates excessively. Root cause: Excessive ambient subtraction gain amplified low-perfusion noise during normal use	FWIA-023 (CPL-100-023) impact assessment for measurement_core on FW-v3.3.8: the change addresses "Excessive ambient subtraction gain amplified low-perfusion noise during normal use" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-023 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-023; verify REQ-MEAS-023; confirm RISK-MEAS-023...	Closed - unable to reproduce after extended test

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-024 CPL-100-024 measurement_core FW-v3.4.2	REQ-MEAS-024 RISK-MEAS-024 TC-MEAS-024	Pulse rate reading jumps randomly. Root cause: Motion rejection filter accepted accelerometer spikes as cardiac peaks during transport	FWIA-024 (CPL-100-024) impact assessment for measurement_core on FW-v3.4.2: the change addresses "Motion rejection filter accepted accelerometer spikes as cardiac peaks during transport" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion...	FWIA-024 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-024; verify REQ-MEAS-024; confirm RISK-MEAS-024...	Closed - confirmed device failure
FWIA-025 CPL-100-025 measurement_core FW-v3.4.9	REQ-MEAS-025 RISK-MEAS-025 TC-MEAS-025	Measurement takes too long to stabilize. Root cause: Stabilization timer restarted repeatedly when perfusion index oscillated near the low-signal boundary	FWIA-025 (CPL-100-025) impact assessment for measurement_core on FW-v3.4.9: the change addresses "Stabilization timer restarted repeatedly when perfusion index oscillated near the low-signal boundary" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization...	FWIA-025 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-025; verify REQ-MEAS-025; confirm RISK-MEAS-025...	Closed - confirmed device failure
FWIA-026 CPL-100-026 measurement_core FW-v3.5.3	REQ-MEAS-026 RISK-MEAS-026 TC-MEAS-026	Device continuously searches for signal. Root cause: Finger-present threshold was raised by a factory calibration coefficient copied from the wrong fixture	FWIA-026 (CPL-100-026) impact assessment for measurement_core on FW-v3.5.3: the change addresses "Finger-present threshold was raised by a factory calibration coefficient copied from the wrong fixture" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization...	FWIA-026 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-026; verify REQ-MEAS-026; confirm RISK-MEAS-026...	Closed - confirmed device failure
FWIA-027 CPL-100-027 measurement_core FW-v3.6.0	REQ-MEAS-027 RISK-MEAS-027 TC-MEAS-027	Finger detection fails. Root cause: Clip optical gap widened enough to reduce received signal below detection margin	FWIA-027 (CPL-100-027) impact assessment for measurement_core on FW-v3.6.0: the change addresses "Clip optical gap widened enough to reduce received signal below detection margin" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-027 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-027; verify REQ-MEAS-027; confirm RISK-MEAS-027...	Closed - confirmed device failure
FWIA-028 CPL-100-028 measurement_core FW-v3.3.4	REQ-MEAS-028 RISK-MEAS-028 TC-MEAS-028	Measurement stops midway. Root cause: Sample FIFO overflowed when display refresh preempted measurement processing	FWIA-028 (CPL-100-028) impact assessment for measurement_core on FW-v3.3.4: the change addresses "Sample FIFO overflowed when display refresh preempted measurement processing" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-028 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-028; verify REQ-MEAS-028; confirm RISK-MEAS-028...	Closed - confirmed device failure
FWIA-029 CPL-100-029 measurement_core FW-v3.4.0	REQ-MEAS-029 RISK-MEAS-029 TC-MEAS-029	Device reports impossible SpO2 values. Root cause: Integer bounds check allowed SpO2 display value above physiologic range after divide-by-small denominator	FWIA-029 (CPL-100-029) impact assessment for measurement_core on FW-v3.4.0: the change addresses "Integer bounds check allowed SpO2 display value above physiologic range after divide-by-small denominator" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization...	FWIA-029 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-029; verify REQ-MEAS-029; confirm RISK-MEAS-029...	Closed - confirmed software defect
FWIA-030 CPL-100-030 measurement_core FW-v3.4.6	REQ-MEAS-030 RISK-MEAS-030 TC-MEAS-030	Pulse waveform is not displayed. Root cause: Waveform renderer was disabled by a feature flag left set during service programming	FWIA-030 (CPL-100-030) impact assessment for measurement_core on FW-v3.4.6: the change addresses "Waveform renderer was disabled by a feature flag left set during service programming" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-030 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-030; verify REQ-MEAS-030; confirm RISK-MEAS-030...	Closed - confirmed device failure

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-031 CPL-100-031 measurement_core FW-v3.5.1	REQ-MEAS-031 RISK-MEAS-031 TC-MEAS-031	Pulse strength indicator is inaccurate. Root cause: Pulse strength scaling table used adult transmission assumptions for low-amplitude pediatric signals	FWIA-031 (CPL-100-031) impact assessment for measurement_core on FW-v3.5.1: the change addresses "Pulse strength scaling table used adult transmission assumptions for low-amplitude pediatric signals" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization...	FWIA-031 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-031; verify REQ-MEAS-031; confirm RISK-MEAS-031...	Closed - confirmed device failure
FWIA-032 CPL-100-032 measurement_core FW-v3.5.7	REQ-MEAS-032 RISK-MEAS-032 TC-MEAS-032	Perfusion index is missing. Root cause: Perfusion index channel was omitted from the telemetry map in firmware 3.4.2	FWIA-032 (CPL-100-032) impact assessment for measurement_core on FW-v3.5.7: the change addresses "Perfusion index channel was omitted from the telemetry map in firmware 3.4.2" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-032 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-032; verify REQ-MEAS-032; confirm RISK-MEAS-032...	Closed - confirmed software defect
FWIA-033 CPL-100-033 measurement_core FW-v3.2.9	REQ-MEAS-033 RISK-MEAS-033 TC-MEAS-033	Signal quality indicator always shows poor. Root cause: Signal-quality algorithm overweighted DC level after a supplier LED bin change	FWIA-033 (CPL-100-033) impact assessment for measurement_core on FW-v3.2.9: the change addresses "Signal-quality algorithm overweighted DC level after a supplier LED bin change" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-033 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-033; verify REQ-MEAS-033; confirm RISK-MEAS-033...	Closed - confirmed device failure
FWIA-034 CPL-100-034 measurement_core FW-v3.3.8	REQ-MEAS-034 RISK-MEAS-034 TC-MEAS-034	Device cannot detect weak pulse. Root cause: Weak-pulse classifier threshold was too conservative for validated low perfusion simulator profiles	FWIA-034 (CPL-100-034) impact assessment for measurement_core on FW-v3.3.8: the change addresses "Weak-pulse classifier threshold was too conservative for validated low perfusion simulator profiles" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing,...	FWIA-034 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-034; verify REQ-MEAS-034; confirm RISK-MEAS-034...	Closed - confirmed device failure
FWIA-035 CPL-100-035 measurement_core FW-v3.4.2	REQ-MEAS-035 RISK-MEAS-035 TC-MEAS-035	Readings differ significantly from clinical device. Root cause: Optical calibration drifted outside tolerance following lens carrier creep	FWIA-035 (CPL-100-035) impact assessment for measurement_core on FW-v3.4.2: the change addresses "Optical calibration drifted outside tolerance following lens carrier creep" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-035 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-035; verify REQ-MEAS-035; confirm RISK-MEAS-035...	Closed - confirmed device failure
FWIA-036 CPL-100-036 measurement_core FW-v3.4.9	REQ-MEAS-036 RISK-MEAS-036 TC-MEAS-036	Measurement accuracy decreases in cold fingers. Root cause: Cold-induced vasoconstriction reduced signal below labeling limits; device operated within specification	FWIA-036 (CPL-100-036) impact assessment for measurement_core on FW-v3.4.9: the change addresses "Cold-induced vasoconstriction reduced signal below labeling limits; device operated within specification" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization...	FWIA-036 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-036; verify REQ-MEAS-036; confirm RISK-MEAS-036...	Closed - use/environmental cause
FWIA-037 CPL-100-037 measurement_core FW-v3.5.3	REQ-MEAS-037 RISK-MEAS-037 TC-MEAS-037	Motion causes measurement failure. Root cause: Motion artifact exceeded validated use conditions because the clip was used during ambulation	FWIA-037 (CPL-100-037) impact assessment for measurement_core on FW-v3.5.3: the change addresses "Motion artifact exceeded validated use conditions because the clip was used during ambulation" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing,...	FWIA-037 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-037; verify REQ-MEAS-037; confirm RISK-MEAS-037...	Closed - use/environmental cause

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-038 CPL-100-038 measurement_core FW-v3.6.0	REQ-MEAS-038 RISK-MEAS-038 TC-MEAS-038	Nail polish causes inaccurate readings. Root cause: Dark nail coating attenuated red and IR channels beyond compensation range	FWIA-038 (CPL-100-038) impact assessment for measurement_core on FW-v3.6.0: the change addresses "Dark nail coating attenuated red and IR channels beyond compensation range" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-038 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-038; verify REQ-MEAS-038; confirm RISK-MEAS-038...	Closed - use/environmental cause
FWIA-039 CPL-100-039 measurement_core FW-v3.3.4	REQ-MEAS-039 RISK-MEAS-039 TC-MEAS-039	Artificial nails prevent detection. Root cause: Artificial nail length prevented full finger insertion and blocked the optical path	FWIA-039 (CPL-100-039) impact assessment for measurement_core on FW-v3.3.4: the change addresses "Artificial nail length prevented full finger insertion and blocked the optical path" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing, perfusion and...	FWIA-039 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-039; verify REQ-MEAS-039; confirm RISK-MEAS-039...	Closed - use/environmental cause
FWIA-040 CPL-100-040 measurement_core FW-v3.4.0	REQ-MEAS-040 RISK-MEAS-040 TC-MEAS-040	Dark skin tone affects accuracy. Root cause: Bias review identified calibration residual at low perfusion/high melanin test condition requiring algorithm update	FWIA-040 (CPL-100-040) impact assessment for measurement_core on FW-v3.4.0: the change addresses "Bias review identified calibration residual at low perfusion/high melanin test condition requiring algorithm update" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits,...	FWIA-040 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-040; verify REQ-MEAS-040; confirm RISK-MEAS-040...	Closed - field correction initiated
FWIA-041 CPL-100-041 measurement_core FW-v3.4.6	REQ-MEAS-041 RISK-MEAS-041 TC-MEAS-041	Bright sunlight interferes with readings. Root cause: Direct sunlight saturated the photodiode despite ambient cancellation due to unshielded outdoor use	FWIA-041 (CPL-100-041) impact assessment for measurement_core on FW-v3.4.6: the change addresses "Direct sunlight saturated the photodiode despite ambient cancellation due to unshielded outdoor use" within the SpO2 and pulse-rate measurement algorithm. Software impact: evaluate signal acquisition, adaptive gain, artifact rejection, plausibility limits, stabilization timing,...	FWIA-041 required verification: perform Unit Testing; Algorithm Dataset Regression; Measurement Accuracy Verification; Boundary Testing; Motion/Low-Perfusion Stress Testing; Integration Testing; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-MEAS-041; verify REQ-MEAS-041; confirm RISK-MEAS-041...	Closed - use/environmental cause
FWIA-042 CPL-100-042 optical_frontend FW-v3.5.1	REQ-SENS-042 RISK-SENS-042 TC-SENS-042	LED sensor fails to illuminate. Root cause: Red LED bond wire lifted after thermal cycling of the optical emitter package	FWIA-042 (CPL-100-042) impact assessment for optical_frontend on FW-v3.5.1: the change addresses "Red LED bond wire lifted after thermal cycling of the optical emitter package" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection, sensor-fault gating, and...	FWIA-042 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-042; verify REQ-SENS-042; confirm RISK-SENS-042...	Closed - confirmed device failure
FWIA-043 CPL-100-043 optical_frontend FW-v3.5.7	REQ-SENS-043 RISK-SENS-043 TC-SENS-043	Infrared LED malfunction. Root cause: IR LED forward voltage drifted high in supplier lot OS-2218, reducing drive current	FWIA-043 (CPL-100-043) impact assessment for optical_frontend on FW-v3.5.7: the change addresses "IR LED forward voltage drifted high in supplier lot OS-2218, reducing drive current" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection, sensor-fault gating,...	FWIA-043 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-043; verify REQ-SENS-043; confirm RISK-SENS-043...	Closed - confirmed supplier component issue
FWIA-044 CPL-100-044 optical_frontend FW-v3.2.9	REQ-SENS-044 RISK-SENS-044 TC-SENS-044	Red LED intensity is inconsistent. Root cause: Red LED current regulator oscillated because output capacitor ESR was outside derating assumption	FWIA-044 (CPL-100-044) impact assessment for optical_frontend on FW-v3.2.9: the change addresses "Red LED current regulator oscillated because output capacitor ESR was outside derating assumption" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection,...	FWIA-044 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-044; verify REQ-SENS-044; confirm RISK-SENS-044...	Closed - confirmed supplier component issue
FWIA-045 CPL-100-045 optical_frontend FW-v3.3.8	REQ-SENS-045 RISK-SENS-045 TC-SENS-045	Photodiode sensor failure. Root cause: Photodiode ESD damage increased junction noise and reduced responsivity	FWIA-045 (CPL-100-045) impact assessment for optical_frontend on FW-v3.3.8: the change addresses "Photodiode ESD damage increased junction noise and reduced responsivity" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection, sensor-fault gating, and...	FWIA-045 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-045; verify REQ-SENS-045; confirm RISK-SENS-045...	Closed - confirmed supplier component issue

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-046 CPL-100-046 optical_frontend FW-v3.4.2	REQ-SENS-046 RISK-SENS-046 TC-SENS-046	Sensor calibration error. Root cause: Factory optical gain calibration was performed with an expired neutral density filter	FWIA-046 (CPL-100-046) impact assessment for optical_frontend on FW-v3.4.2: the change addresses "Factory optical gain calibration was performed with an expired neutral density filter" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection, sensor-fault gating,....	FWIA-046 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-046; verify REQ-SENS-046; confirm RISK-SENS-046...	Closed - confirmed device failure
FWIA-047 CPL-100-047 optical_frontend FW-v3.4.9	REQ-SENS-047 RISK-SENS-047 TC-SENS-047	Optical sensor contamination detected. Root cause: Finger well contained dried hand-sanitizer film over the optical window	FWIA-047 (CPL-100-047) impact assessment for optical_frontend on FW-v3.4.9: the change addresses "Finger well contained dried hand-sanitizer film over the optical window" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection, sensor-fault gating, and...	FWIA-047 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-047; verify REQ-SENS-047; confirm RISK-SENS-047...	Closed - confirmed device failure
FWIA-048 CPL-100-048 optical_frontend FW-v3.5.3	REQ-SENS-048 RISK-SENS-048 TC-SENS-048	Finger clip spring loses tension. Root cause: Clip spring lost preload after hinge pin hole ovalization from repeated opening beyond stop	FWIA-048 (CPL-100-048) impact assessment for optical_frontend on FW-v3.5.3: the change addresses "Clip spring lost preload after hinge pin hole ovalization from repeated opening beyond stop" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection, sensor-fault...	FWIA-048 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-048; verify REQ-SENS-048; confirm RISK-SENS-048...	Closed - confirmed supplier component issue
FWIA-049 CPL-100-049 optical_frontend FW-v3.6.0	REQ-SENS-049 RISK-SENS-049 TC-SENS-049	Sensor housing becomes loose. Root cause: Ultrasonic weld energy was low on upper housing, allowing sensor pod movement	FWIA-049 (CPL-100-049) impact assessment for optical_frontend on FW-v3.6.0: the change addresses "Ultrasonic weld energy was low on upper housing, allowing sensor pod movement" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection, sensor-fault gating, and...	FWIA-049 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-049; verify REQ-SENS-049; confirm RISK-SENS-049...	Closed - confirmed supplier component issue
FWIA-050 CPL-100-050 optical_frontend FW-v3.3.4	REQ-SENS-050 RISK-SENS-050 TC-SENS-050	Sensor cable connection failure. Root cause: Micro-coax shield termination cracked at strain-relief transition	FWIA-050 (CPL-100-050) impact assessment for optical_frontend on FW-v3.3.4: the change addresses "Micro-coax shield termination cracked at strain-relief transition" within the optical sensor frontend firmware. Software impact: evaluate LED drive control, photodiode range checks, calibration constant handling, contamination detection, sensor-fault gating, and cable/clip...	FWIA-050 required verification: perform Unit Testing; Hardware-in-the-Loop Integration Testing; Sensor Fault Injection; Calibration CRC Verification; Measurement Inhibit Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SENS-050; verify REQ-SENS-050; confirm RISK-SENS-050...	Closed - confirmed device failure
FWIA-051 CPL-100-051 alarm_manager FW-v3.4.0	REQ-ALRM-051 RISK-ALRM-051 TC-ALRM-051	High SpO2 alarm does not trigger. Root cause: High SpO2 alarm comparator used inclusive threshold on falling edge only	FWIA-051 (CPL-100-051) impact assessment for alarm_manager on FW-v3.4.0: the change addresses "High SpO2 alarm comparator used inclusive threshold on falling edge only" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers, clearing logic, and...	FWIA-051 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-051; verify REQ-ALRM-051; confirm RISK-ALRM-051 risk-control...	Closed - field correction initiated
FWIA-052 CPL-100-052 alarm_manager FW-v3.4.6	REQ-ALRM-052 RISK-ALRM-052 TC-ALRM-052	Low SpO2 alarm does not trigger. Root cause: Low SpO2 alarm inhibit timer was not cleared after transient sensor-off condition	FWIA-052 (CPL-100-052) impact assessment for alarm_manager on FW-v3.4.6: the change addresses "Low SpO2 alarm inhibit timer was not cleared after transient sensor-off condition" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers, clearing...	FWIA-052 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-052; verify REQ-ALRM-052; confirm RISK-ALRM-052 risk-control...	Closed - field correction initiated
FWIA-053 CPL-100-053 alarm_manager FW-v3.5.1	REQ-ALRM-053 RISK-ALRM-053 TC-ALRM-053	High pulse rate alarm missing. Root cause: High pulse alarm event was queued behind noncritical BLE sync notifications	FWIA-053 (CPL-100-053) impact assessment for alarm_manager on FW-v3.5.1: the change addresses "High pulse alarm event was queued behind noncritical BLE sync notifications" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers, clearing logic, and...	FWIA-053 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-053; verify REQ-ALRM-053; confirm RISK-ALRM-053 risk-control...	Closed - confirmed device failure

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-054 CPL-100-054 alarm_manager FW-v3.5.7	REQ-ALRM-054 RISK-ALRM-054 TC-ALRM-054	Low pulse rate alarm missing. Root cause: Low pulse alarm limit was stored in bpm but compared as packed BCD after restore	FWIA-054 (CPL-100-054) impact assessment for alarm_manager on FW-v3.5.7: the change addresses "Low pulse alarm limit was stored in bpm but compared as packed BCD after restore" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers, clearing logic,...	FWIA-054 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-054; verify REQ-ALRM-054; confirm RISK-ALRM-054 risk-control...	Closed - confirmed device failure
FWIA-055 CPL-100-055 alarm_manager FW-v3.2.9	REQ-ALRM-055 RISK-ALRM-055 TC-ALRM-055	Alarm sounds continuously without reason. Root cause: Alarm tone driver received repeated false alarm requests from noisy signal-quality state	FWIA-055 (CPL-100-055) impact assessment for alarm_manager on FW-v3.2.9: the change addresses "Alarm tone driver received repeated false alarm requests from noisy signal-quality state" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers,...	FWIA-055 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-055; verify REQ-ALRM-055; confirm RISK-ALRM-055 risk-control...	Closed - confirmed device failure
FWIA-056 CPL-100-056 alarm_manager FW-v3.3.8	REQ-ALRM-056 RISK-ALRM-056 TC-ALRM-056	Alarm volume cannot be adjusted. Root cause: Volume setting potentiometer ADC channel saturated because reference divider solder joint was cracked	FWIA-056 (CPL-100-056) impact assessment for alarm_manager on FW-v3.3.8: the change addresses "Volume setting potentiometer ADC channel saturated because reference divider solder joint was cracked" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay...	FWIA-056 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-056; verify REQ-ALRM-056; confirm RISK-ALRM-056 risk-control...	Closed - confirmed device failure
FWIA-057 CPL-100-057 alarm_manager FW-v3.4.2	REQ-ALRM-057 RISK-ALRM-057 TC-ALRM-057	Alarm remains muted permanently. Root cause: Mute-latch bit remained set after failed acknowledgement write to alarm state EEPROM	FWIA-057 (CPL-100-057) impact assessment for alarm_manager on FW-v3.4.2: the change addresses "Mute-latch bit remained set after failed acknowledgement write to alarm state EEPROM" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers, clearing...	FWIA-057 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-057; verify REQ-ALRM-057; confirm RISK-ALRM-057 risk-control...	Closed - confirmed software defect
FWIA-058 CPL-100-058 alarm_manager FW-v3.4.9	REQ-ALRM-058 RISK-ALRM-058 TC-ALRM-058	Alarm delay exceeds specification. Root cause: Alarm annunciation latency exceeded requirement under simultaneous waveform, log, and BLE load	FWIA-058 (CPL-100-058) impact assessment for alarm_manager on FW-v3.4.9: the change addresses "Alarm annunciation latency exceeded requirement under simultaneous waveform, log, and BLE load" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers,...	FWIA-058 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-058; verify REQ-ALRM-058; confirm RISK-ALRM-058 risk-control...	Closed - field correction initiated
FWIA-059 CPL-100-059 alarm_manager FW-v3.5.3	REQ-ALRM-059 RISK-ALRM-059 TC-ALRM-059	Alarm icon does not appear. Root cause: Alarm icon asset pointer referenced the wrong display page after theme update	FWIA-059 (CPL-100-059) impact assessment for alarm_manager on FW-v3.5.3: the change addresses "Alarm icon asset pointer referenced the wrong display page after theme update" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers, clearing logic,...	FWIA-059 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-059; verify REQ-ALRM-059; confirm RISK-ALRM-059 risk-control...	Closed - confirmed device failure
FWIA-060 CPL-100-060 alarm_manager FW-v3.6.0	REQ-ALRM-060 RISK-ALRM-060 TC-ALRM-060	Alarm continues after condition is cleared. Root cause: Alarm clear debounce was too long for rapidly recovering simulated physiologic events	FWIA-060 (CPL-100-060) impact assessment for alarm_manager on FW-v3.6.0: the change addresses "Alarm clear debounce was too long for rapidly recovering simulated physiologic events" within the physiological and technical alarm firmware. Software impact: evaluate alarm-limit evaluation, priority mapping, audible/visual annunciation, silence behavior, delay timers, clearing...	FWIA-060 required verification: perform Unit Testing; IEC 60601-1-8 Alarm Verification; Alarm Timing Boundary Testing; Audible/Visual Integration Testing; Mute/Volume Regression; Risk Control Verification; System Testing; Traceability Review. Traceability: execute TC-ALRM-060; verify REQ-ALRM-060; confirm RISK-ALRM-060 risk-control...	Closed - confirmed software defect
FWIA-061 CPL-100-061 storage_manager FW-v3.3.4	REQ-DATA-061 RISK-DATA-061 TC-DATA-061	Date and time reset after power cycle. Root cause: RTC backup capacitor was installed reversed on one assembly lot	FWIA-061 (CPL-100-061) impact assessment for storage_manager on FW-v3.3.4: the change addresses "RTC backup capacitor was installed reversed on one assembly lot" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage, factory reset, measurement-history management,...	FWIA-061 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-061; verify REQ-DATA-061; confirm RISK-DATA-061 risk-control effectiveness and...	Closed - confirmed device failure

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-062 CPL-100-062 storage_manager FW-v3.4.0	REQ-DATA-062 RISK-DATA-062 TC-DATA-062	Settings are not saved. Root cause: Settings commit routine returned success before flash write completion	FWIA-062 (CPL-100-062) impact assessment for storage_manager on FW-v3.4.0: the change addresses "Settings commit routine returned success before flash write completion" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage, factory reset, measurement-history...	FWIA-062 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-062; verify REQ-DATA-062; confirm RISK-DATA-062 risk-control effectiveness and...	Closed - confirmed device failure
FWIA-063 CPL-100-063 storage_manager FW-v3.4.6	REQ-DATA-063 RISK-DATA-063 TC-DATA-063	Factory reset does not work. Root cause: Factory-reset command handler skipped user partition erase when language resource pack was present	FWIA-063 (CPL-100-063) impact assessment for storage_manager on FW-v3.4.6: the change addresses "Factory-reset command handler skipped user partition erase when language resource pack was present" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage, factory...	FWIA-063 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-063; verify REQ-DATA-063; confirm RISK-DATA-063 risk-control effectiveness and...	Closed - confirmed software defect
FWIA-064 CPL-100-064 storage_manager FW-v3.5.1	REQ-DATA-064 RISK-DATA-064 TC-DATA-064	Device language cannot be changed. Root cause: Locale table lacked the requested language pack for the regional SKU shipped to customer	FWIA-064 (CPL-100-064) impact assessment for storage_manager on FW-v3.5.1: the change addresses "Locale table lacked the requested language pack for the regional SKU shipped to customer" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage, factory reset,...	FWIA-064 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-064; verify REQ-DATA-064; confirm RISK-DATA-064 risk-control effectiveness and...	Closed - no fault found
FWIA-065 CPL-100-065 storage_manager FW-v3.5.7	REQ-DATA-065 RISK-DATA-065 TC-DATA-065	User profile cannot be created. Root cause: Profile database rejected creation after duplicate UUID generated during interrupted setup	FWIA-065 (CPL-100-065) impact assessment for storage_manager on FW-v3.5.7: the change addresses "Profile database rejected creation after duplicate UUID generated during interrupted setup" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage, factory reset,...	FWIA-065 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-065; verify REQ-DATA-065; confirm RISK-DATA-065 risk-control effectiveness and...	Closed - confirmed device failure
FWIA-066 CPL-100-066 storage_manager FW-v3.2.9	REQ-DATA-066 RISK-DATA-066 TC-DATA-066	Memory storage is full unexpectedly. Root cause: Circular log index failed to wrap, reporting storage full with unused sectors available	FWIA-066 (CPL-100-066) impact assessment for storage_manager on FW-v3.2.9: the change addresses "Circular log index failed to wrap, reporting storage full with unused sectors available" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage, factory reset,...	FWIA-066 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-066; verify REQ-DATA-066; confirm RISK-DATA-066 risk-control effectiveness and...	Closed - confirmed software defect
FWIA-067 CPL-100-067 storage_manager FW-v3.3.8	REQ-DATA-067 RISK-DATA-067 TC-DATA-067	Stored measurements disappear. Root cause: History compaction routine deleted recent records when daylight-saving transition created duplicate keys	FWIA-067 (CPL-100-067) impact assessment for storage_manager on FW-v3.3.8: the change addresses "History compaction routine deleted recent records when daylight-saving transition created duplicate keys" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage,...	FWIA-067 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-067; verify REQ-DATA-067; confirm RISK-DATA-067 risk-control effectiveness and...	Closed - confirmed software defect
FWIA-068 CPL-100-068 storage_manager FW-v3.4.2	REQ-DATA-068 RISK-DATA-068 TC-DATA-068	Measurement history displays incorrect timestamps. Root cause: Time-zone offset was applied twice during history rendering after phone sync	FWIA-068 (CPL-100-068) impact assessment for storage_manager on FW-v3.4.2: the change addresses "Time-zone offset was applied twice during history rendering after phone sync" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage, factory reset, measurement-history...	FWIA-068 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-068; verify REQ-DATA-068; confirm RISK-DATA-068 risk-control effectiveness and...	Closed - confirmed software defect
FWIA-069 CPL-100-069 storage_manager FW-v3.4.9	REQ-DATA-069 RISK-DATA-069 TC-DATA-069	Data export fails. Root cause: CSV export failed on Windows hosts because USB descriptor reported an unsupported composite subclass	FWIA-069 (CPL-100-069) impact assessment for storage_manager on FW-v3.4.9: the change addresses "CSV export failed on Windows hosts because USB descriptor reported an unsupported composite subclass" within the data integrity, settings, history, and export firmware. Software impact: evaluate nonvolatile settings commit, RTC retention, profile and language storage, factory...	FWIA-069 required verification: perform Unit Testing; Data Integrity Verification; Power-Cycle and Brownout Testing; Memory-Full Boundary Testing; Export Validation; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-DATA-069; verify REQ-DATA-069; confirm RISK-DATA-069 risk-control effectiveness and...	Closed - confirmed device failure

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-070 CPL-100-070 connectivity_stack FW-v3.5.3	REQ-CONN-070 RISK-CONN-070 TC-CONN-070	USB connection is not recognized. Root cause: USB-C receptacle shell was mechanically deformed, preventing full cable insertion	FWIA-070 (CPL-100-070) impact assessment for connectivity_stack on FW-v3.5.3: the change addresses "USB-C receptacle shell was mechanically deformed, preventing full cable insertion" within the USB, Bluetooth, Wi-Fi, mobile application, and cloud communication firmware. Software impact: evaluate device enumeration, pairing state machine, reconnect handling, sync queue,...	FWIA-070 required verification: perform Unit Testing; Protocol Integration Testing; Cybersecurity/Authentication Review; Interrupted Transfer Stress Testing; Data Integrity Verification; System Testing; Regression Testing; Traceability Review. Traceability: execute TC-CONN-070; verify REQ-CONN-070; confirm RISK-CONN-070 risk-control...	Closed - confirmed device failure
FWIA-071 CPL-100-071 connectivity_stack FW-v3.6.0	REQ-CONN-071 RISK-CONN-071 TC-CONN-071	Bluetooth pairing fails. Root cause: BLE bonding table filled with stale phone entries and blocked new pairing	FWIA-071 (CPL-100-071) impact assessment for connectivity_stack on FW-v3.6.0: the change addresses "BLE bonding table filled with stale phone entries and blocked new pairing" within the USB, Bluetooth, Wi-Fi, mobile application, and cloud communication firmware. Software impact: evaluate device enumeration, pairing state machine, reconnect handling, sync queue,...	FWIA-071 required verification: perform Unit Testing; Protocol Integration Testing; Cybersecurity/Authentication Review; Interrupted Transfer Stress Testing; Data Integrity Verification; System Testing; Regression Testing; Traceability Review. Traceability: execute TC-CONN-071; verify REQ-CONN-071; confirm RISK-CONN-071 risk-control...	Closed - confirmed device failure
FWIA-072 CPL-100-072 connectivity_stack FW-v3.3.4	REQ-CONN-072 RISK-CONN-072 TC-CONN-072	Bluetooth disconnects frequently. Root cause: BLE connection interval request conflicted with Android 14 peripheral policy	FWIA-072 (CPL-100-072) impact assessment for connectivity_stack on FW-v3.3.4: the change addresses "BLE connection interval request conflicted with Android 14 peripheral policy" within the USB, Bluetooth, Wi-Fi, mobile application, and cloud communication firmware. Software impact: evaluate device enumeration, pairing state machine, reconnect handling, sync queue,...	FWIA-072 required verification: perform Unit Testing; Protocol Integration Testing; Cybersecurity/Authentication Review; Interrupted Transfer Stress Testing; Data Integrity Verification; System Testing; Regression Testing; Traceability Review. Traceability: execute TC-CONN-072; verify REQ-CONN-072; confirm RISK-CONN-072 risk-control...	Closed - unable to reproduce after extended test
FWIA-073 CPL-100-073 connectivity_stack FW-v3.4.0	REQ-CONN-073 RISK-CONN-073 TC-CONN-073	Mobile app cannot sync data. Root cause: Mobile app sync parser rejected device records containing null perfusion-index fields	FWIA-073 (CPL-100-073) impact assessment for connectivity_stack on FW-v3.4.0: the change addresses "Mobile app sync parser rejected device records containing null perfusion-index fields" within the USB, Bluetooth, Wi-Fi, mobile application, and cloud communication firmware. Software impact: evaluate device enumeration, pairing state machine, reconnect handling, sync queue,...	FWIA-073 required verification: perform Unit Testing; Protocol Integration Testing; Cybersecurity/Authentication Review; Interrupted Transfer Stress Testing; Data Integrity Verification; System Testing; Regression Testing; Traceability Review. Traceability: execute TC-CONN-073; verify REQ-CONN-073; confirm RISK-CONN-073 risk-control...	Closed - confirmed software defect
FWIA-074 CPL-100-074 connectivity_stack FW-v3.4.6	REQ-CONN-074 RISK-CONN-074 TC-CONN-074	Wi-Fi connection fails. Root cause: Wi-Fi provisioning failed because 5 GHz SSIDs were shown although device supports 2.4 GHz only	FWIA-074 (CPL-100-074) impact assessment for connectivity_stack on FW-v3.4.6: the change addresses "Wi-Fi provisioning failed because 5 GHz SSIDs were shown although device supports 2.4 GHz only" within the USB, Bluetooth, Wi-Fi, mobile application, and cloud communication firmware. Software impact: evaluate device enumeration, pairing state machine, reconnect handling,...	FWIA-074 required verification: perform Unit Testing; Protocol Integration Testing; Cybersecurity/Authentication Review; Interrupted Transfer Stress Testing; Data Integrity Verification; System Testing; Regression Testing; Traceability Review. Traceability: execute TC-CONN-074; verify REQ-CONN-074; confirm RISK-CONN-074 risk-control...	Closed - confirmed device failure
FWIA-075 CPL-100-075 connectivity_stack FW-v3.5.1	REQ-CONN-075 RISK-CONN-075 TC-CONN-075	Cloud upload unsuccessful. Root cause: Cloud upload token expired during long offline queue replay and was not refreshed	FWIA-075 (CPL-100-075) impact assessment for connectivity_stack on FW-v3.5.1: the change addresses "Cloud upload token expired during long offline queue replay and was not refreshed" within the USB, Bluetooth, Wi-Fi, mobile application, and cloud communication firmware. Software impact: evaluate device enumeration, pairing state machine, reconnect handling, sync queue,...	FWIA-075 required verification: perform Unit Testing; Protocol Integration Testing; Cybersecurity/Authentication Review; Interrupted Transfer Stress Testing; Data Integrity Verification; System Testing; Regression Testing; Traceability Review. Traceability: execute TC-CONN-075; verify REQ-CONN-075; confirm RISK-CONN-075 risk-control...	Closed - confirmed software defect
FWIA-076 CPL-100-076 update_identity_manager FW-v3.5.7	REQ-FWID-076 RISK-FWID-076 TC-FWID-076	Firmware update fails. Root cause: Firmware package signature verification failed after interrupted download left a partial manifest	FWIA-076 (CPL-100-076) impact assessment for update_identity_manager on FW-v3.5.7: the change addresses "Firmware package signature verification failed after interrupted download left a partial manifest" within the firmware update, bootloader, identity, discovery, and error-code firmware. Software impact: evaluate package validation, bootloader transition, rollback/recovery...	FWIA-076 required verification: perform Unit Testing; Firmware Update Validation; Rollback/Recovery Testing; Cybersecurity Signature Verification; Identity/Version Verification; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-FWID-076; verify REQ-FWID-076; confirm RISK-FWID-076 risk-control...	Closed - confirmed software defect

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-077 CPL-100-077 update_identity_manager FW-v3.2.9	REQ-FWID-077 RISK-FWID-077 TC-FWID-077	Firmware update causes boot loop. Root cause: Bootloader rollback slot header was erased before new image validity was confirmed	FWIA-077 (CPL-100-077) impact assessment for update_identity_manager on FW-v3.2.9: the change addresses "Bootloader rollback slot header was erased before new image validity was confirmed" within the firmware update, bootloader, identity, discovery, and error-code firmware. Software impact: evaluate package validation, bootloader transition, rollback/recovery image...	FWIA-077 required verification: perform Unit Testing; Firmware Update Validation; Rollback/Recovery Testing; Cybersecurity Signature Verification; Identity/Version Verification; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-FWID-077; verify REQ-FWID-077; confirm RISK-FWID-077 risk-control...	Closed - field correction initiated
FWIA-078 CPL-100-078 update_identity_manager FW-v3.3.8	REQ-FWID-078 RISK-FWID-078 TC-FWID-078	Device version information is incorrect. Root cause: About-screen version string used application image metadata rather than released package metadata	FWIA-078 (CPL-100-078) impact assessment for update_identity_manager on FW-v3.3.8: the change addresses "About-screen version string used application image metadata rather than released package metadata" within the firmware update, bootloader, identity, discovery, and error-code firmware. Software impact: evaluate package validation, bootloader transition, rollback/recovery...	FWIA-078 required verification: perform Unit Testing; Firmware Update Validation; Rollback/Recovery Testing; Cybersecurity Signature Verification; Identity/Version Verification; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-FWID-078; verify REQ-FWID-078; confirm RISK-FWID-078 risk-control...	Closed - no fault found
FWIA-079 CPL-100-079 update_identity_manager FW-v3.4.2	REQ-FWID-079 RISK-FWID-079 TC-FWID-079	Serial number is missing. Root cause: Serial-number EEPROM field was blank due to production programmer timeout	FWIA-079 (CPL-100-079) impact assessment for update_identity_manager on FW-v3.4.2: the change addresses "Serial-number EEPROM field was blank due to production programmer timeout" within the firmware update, bootloader, identity, discovery, and error-code firmware. Software impact: evaluate package validation, bootloader transition, rollback/recovery image selection,...	FWIA-079 required verification: perform Unit Testing; Firmware Update Validation; Rollback/Recovery Testing; Cybersecurity Signature Verification; Identity/Version Verification; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-FWID-079; verify REQ-FWID-079; confirm RISK-FWID-079 risk-control...	Closed - confirmed device failure
FWIA-080 CPL-100-080 update_identity_manager FW-v3.4.9	REQ-FWID-080 RISK-FWID-080 TC-FWID-080	Device cannot be discovered by companion app. Root cause: Advertising name exceeded BLE payload after localized model suffix was appended	FWIA-080 (CPL-100-080) impact assessment for update_identity_manager on FW-v3.4.9: the change addresses "Advertising name exceeded BLE payload after localized model suffix was appended" within the firmware update, bootloader, identity, discovery, and error-code firmware. Software impact: evaluate package validation, bootloader transition, rollback/recovery image selection,...	FWIA-080 required verification: perform Unit Testing; Firmware Update Validation; Rollback/Recovery Testing; Cybersecurity Signature Verification; Identity/Version Verification; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-FWID-080; verify REQ-FWID-080; confirm RISK-FWID-080 risk-control...	Closed - confirmed software defect
FWIA-081 CPL-100-081 update_identity_manager FW-v3.5.3	REQ-FWID-081 RISK-FWID-081 TC-FWID-081	Error code appears without explanation. Root cause: Error-code lookup table missing entry for optical AFE timeout	FWIA-081 (CPL-100-081) impact assessment for update_identity_manager on FW-v3.5.3: the change addresses "Error-code lookup table missing entry for optical AFE timeout" within the firmware update, bootloader, identity, discovery, and error-code firmware. Software impact: evaluate package validation, bootloader transition, rollback/recovery image selection, firmware/serial...	FWIA-081 required verification: perform Unit Testing; Firmware Update Validation; Rollback/Recovery Testing; Cybersecurity Signature Verification; Identity/Version Verification; Integration Testing; Regression Testing; Traceability Review. Traceability: execute TC-FWID-081; verify REQ-FWID-081; confirm RISK-FWID-081 risk-control...	Closed - no fault found
FWIA-082 CPL-100-082 system_supervisor FW-v3.6.0	REQ-SYS-082 RISK-SYS-082 TC-SYS-082	Device remains stuck on startup logo. Root cause: Boot splash remained active because main task blocked waiting for storage mount retry	FWIA-082 (CPL-100-082) impact assessment for system_supervisor on FW-v3.6.0: the change addresses "Boot splash remained active because main task blocked waiting for storage mount retry" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation,...	FWIA-082 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-082; verify REQ-SYS-082; confirm...	Closed - confirmed software defect
FWIA-083 CPL-100-083 system_supervisor FW-v3.3.4	REQ-SYS-083 RISK-SYS-083 TC-SYS-083	Self-test fails during boot. Root cause: Power-on self-test failed due to marginal crystal startup time at low temperature	FWIA-083 (CPL-100-083) impact assessment for system_supervisor on FW-v3.3.4: the change addresses "Power-on self-test failed due to marginal crystal startup time at low temperature" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation,...	FWIA-083 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-083; verify REQ-SYS-083; confirm...	Closed - confirmed device failure
FWIA-084 CPL-100-084 system_supervisor FW-v3.4.0	REQ-SYS-084 RISK-SYS-084 TC-SYS-084	Internal memory corruption detected. Root cause: Flash wear-leveling metadata failed CRC after repeated incomplete shutdowns	FWIA-084 (CPL-100-084) impact assessment for system_supervisor on FW-v3.4.0: the change addresses "Flash wear-leveling metadata failed CRC after repeated incomplete shutdowns" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation,...	FWIA-084 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-084; verify REQ-SYS-084; confirm...	Closed - field correction initiated

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-085 CPL-100-085 system_supervisor FW-v3.4.6	REQ-SYS-085 RISK-SYS-085 TC-SYS-085	Unexpected firmware exception occurs. Root cause: Null pointer exception in alarm-history serializer after empty log compaction	FWIA-085 (CPL-100-085) impact assessment for system_supervisor on FW-v3.4.6: the change addresses "Null pointer exception in alarm-history serializer after empty log compaction" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation,...	FWIA-085 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-085; verify REQ-SYS-085; confirm...	Closed - confirmed software defect
FWIA-086 CPL-100-086 system_supervisor FW-v3.5.1	REQ-SYS-086 RISK-SYS-086 TC-SYS-086	Device becomes slow after prolonged use. Root cause: Heap fragmentation accumulated after repeated BLE reconnect cycles	FWIA-086 (CPL-100-086) impact assessment for system_supervisor on FW-v3.5.1: the change addresses "Heap fragmentation accumulated after repeated BLE reconnect cycles" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation, recovery-mode...	FWIA-086 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-086; verify REQ-SYS-086; confirm...	Closed - unable to reproduce after extended test
FWIA-087 CPL-100-087 system_supervisor FW-v3.5.7	REQ-SYS-087 RISK-SYS-087 TC-SYS-087	Random system hang during measurement. Root cause: Measurement task deadlocked with storage mutex during automatic log rotation	FWIA-087 (CPL-100-087) impact assessment for system_supervisor on FW-v3.5.7: the change addresses "Measurement task deadlocked with storage mutex during automatic log rotation" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation,...	FWIA-087 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-087; verify REQ-SYS-087; confirm...	Closed - unable to reproduce after extended test
FWIA-088 CPL-100-088 system_supervisor FW-v3.2.9	REQ-SYS-088 RISK-SYS-088 TC-SYS-088	Configuration file becomes corrupted. Root cause: Configuration schema migration wrote mixed-version keys after downgrade/upgrade sequence	FWIA-088 (CPL-100-088) impact assessment for system_supervisor on FW-v3.2.9: the change addresses "Configuration schema migration wrote mixed-version keys after downgrade/upgrade sequence" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation,...	FWIA-088 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-088; verify REQ-SYS-088; confirm...	Closed - confirmed software defect
FWIA-089 CPL-100-089 system_supervisor FW-v3.3.8	REQ-SYS-089 RISK-SYS-089 TC-SYS-089	Device enters recovery mode unexpectedly. Root cause: Recovery-mode flag was asserted by boot counter not reset after a brownout	FWIA-089 (CPL-100-089) impact assessment for system_supervisor on FW-v3.3.8: the change addresses "Recovery-mode flag was asserted by boot counter not reset after a brownout" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation,...	FWIA-089 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-089; verify REQ-SYS-089; confirm...	Closed - confirmed software defect
FWIA-090 CPL-100-090 system_supervisor FW-v3.4.2	REQ-SYS-090 RISK-SYS-090 TC-SYS-090	Calibration data is lost after update. Root cause: Calibration partition was excluded from backup list during firmware update migration	FWIA-090 (CPL-100-090) impact assessment for system_supervisor on FW-v3.4.2: the change addresses "Calibration partition was excluded from backup list during firmware update migration" within the system supervision and fault-handling firmware. Software impact: evaluate POST/self-test, memory integrity checks, exception handling, watchdog strategy, configuration validation,...	FWIA-090 required verification: perform Unit Testing; Startup/Self-Test Verification; Fault Injection; Watchdog and Recovery Testing; Long-Duration Performance Testing; Memory Corruption Stress Testing; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-SYS-090; verify REQ-SYS-090; confirm...	Closed - field correction initiated
FWIA-091 CPL-100-091 mechanical_usability FW-v3.4.9	REQ-USE-091 RISK-USE-091 TC-USE-091	Finger insertion guide is unclear. Root cause: IFU insertion graphic was ambiguous for short fingers and led to shallow placement	FWIA-091 (CPL-100-091) impact assessment for mechanical_usability on FW-v3.4.9: the change addresses "IFU insertion graphic was ambiguous for short fingers and led to shallow placement" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection, poor-signal prompting,...	FWIA-091 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-091; verify REQ-USE-091; confirm RISK-USE-091...	Closed - use/environmental cause
FWIA-092 CPL-100-092 mechanical_usability FW-v3.5.3	REQ-USE-092 RISK-USE-092 TC-USE-092	Device is uncomfortable for pediatric patients. Root cause: Pediatric accessory clip was not included with a kit used below labeled finger-width range	FWIA-092 (CPL-100-092) impact assessment for mechanical_usability on FW-v3.5.3: the change addresses "Pediatric accessory clip was not included with a kit used below labeled finger-width range" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection, poor-signal...	FWIA-092 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-092; verify REQ-USE-092; confirm RISK-USE-092...	Closed - use/environmental cause

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-093 CPL-100-093 mechanical_usability FW-v3.6.0	REQ-USE-093 RISK-USE-093 TC-USE-093	Large fingers do not fit properly. Root cause: Large finger compression reduced arterial pulsatility and caused discomfort	FWIA-093 (CPL-100-093) impact assessment for mechanical_usability on FW-v3.6.0: the change addresses "Large finger compression reduced arterial pulsatility and caused discomfort" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection, poor-signal prompting,...	FWIA-093 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-093; verify REQ-USE-093; confirm RISK-USE-093...	Closed - use/environmental cause
FWIA-094 CPL-100-094 mechanical_usability FW-v3.3.4	REQ-USE-094 RISK-USE-094 TC-USE-094	Small fingers cannot be detected reliably. Root cause: Small finger did not maintain optical alignment because foam pad compression set exceeded limit	FWIA-094 (CPL-100-094) impact assessment for mechanical_usability on FW-v3.3.4: the change addresses "Small finger did not maintain optical alignment because foam pad compression set exceeded limit" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection, poor-signal...	FWIA-094 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-094; verify REQ-USE-094; confirm RISK-USE-094...	Closed - use/environmental cause
FWIA-095 CPL-100-095 mechanical_usability FW-v3.4.0	REQ-USE-095 RISK-USE-095 TC-USE-095	Device emits unusual clicking sound. Root cause: Hinge stop clicked against housing rib after tolerance stack-up in lower shell	FWIA-095 (CPL-100-095) impact assessment for mechanical_usability on FW-v3.4.0: the change addresses "Hinge stop clicked against housing rib after tolerance stack-up in lower shell" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection, poor-signal prompting,...	FWIA-095 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-095; verify REQ-USE-095; confirm RISK-USE-095...	Closed - no fault found
FWIA-096 CPL-100-096 mechanical_usability FW-v3.4.6	REQ-USE-096 RISK-USE-096 TC-USE-096	Plastic casing develops cracks. Root cause: Polycarbonate housing cracked after repeated exposure to quaternary ammonium cleaner not listed for use	FWIA-096 (CPL-100-096) impact assessment for mechanical_usability on FW-v3.4.6: the change addresses "Polycarbonate housing cracked after repeated exposure to quaternary ammonium cleaner not listed for use" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection,...	FWIA-096 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-096; verify REQ-USE-096; confirm RISK-USE-096...	Closed - confirmed device failure
FWIA-097 CPL-100-097 mechanical_usability FW-v3.5.1	REQ-USE-097 RISK-USE-097 TC-USE-097	Water splash causes malfunction. Root cause: Liquid ingress reached the side-key board through damaged USB port cover	FWIA-097 (CPL-100-097) impact assessment for mechanical_usability on FW-v3.5.1: the change addresses "Liquid ingress reached the side-key board through damaged USB port cover" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection, poor-signal prompting,...	FWIA-097 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-097; verify REQ-USE-097; confirm RISK-USE-097...	Closed - use/environmental cause
FWIA-098 CPL-100-098 mechanical_usability FW-v3.5.7	REQ-USE-098 RISK-USE-098 TC-USE-098	Dust accumulation affects sensor performance. Root cause: Dust collected behind the optical window in a high-particulate home environment	FWIA-098 (CPL-100-098) impact assessment for mechanical_usability on FW-v3.5.7: the change addresses "Dust collected behind the optical window in a high-particulate home environment" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection, poor-signal prompting,...	FWIA-098 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-098; verify REQ-USE-098; confirm RISK-USE-098...	Closed - use/environmental cause
FWIA-099 CPL-100-099 mechanical_usability FW-v3.2.9	REQ-USE-099 RISK-USE-099 TC-USE-099	Device fails after accidental drop. Root cause: Drop impact fractured the main PCB near the battery connector solder joints	FWIA-099 (CPL-100-099) impact assessment for mechanical_usability on FW-v3.2.9: the change addresses "Drop impact fractured the main PCB near the battery connector solder joints" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection, poor-signal prompting,...	FWIA-099 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-099; verify REQ-USE-099; confirm RISK-USE-099...	Closed - field correction initiated

Change / Module	Requirement / Risk / Test	Resolved Issue	Impact Summary	Required Verification	Release Decision
FWIA-100 CPL-100-100 mechanical_usability FW-v3.3.8	REQ-USE-100 RISK-USE-100 TC-USE-100	Measurement results are inconsistent across repeated tests under identical conditions. Root cause: Combination of loose clip hinge and borderline optical calibration caused repeatability outside acceptance	FWIA-100 (CPL-100-100) impact assessment for mechanical_usability on FW-v3.3.8: the change addresses "Combination of loose clip hinge and borderline optical calibration caused repeatability outside acceptance" within the use-condition, placement-detection, and mechanical-interface support firmware. Software impact: evaluate finger-placement guidance, sensor-fit detection,...	FWIA-100 required verification: perform Usability Validation; System Testing; Mechanical/Environmental Regression; Placement Detection Boundary Testing; Measurement Repeatability Verification; Risk Control Verification; Regression Testing; Traceability Review. Traceability: execute TC-USE-100; verify REQ-USE-100; confirm RISK-USE-100...	Closed - confirmed device failure

Approval Section

Role	Name / Title	Signature	Date
Prepared by	Senior Medical Device Software Engineer		
Reviewed by	Software Quality / Systems Engineering		
Approved by	Quality Assurance / Regulatory Affairs		
Release authorized by	Design Authority		